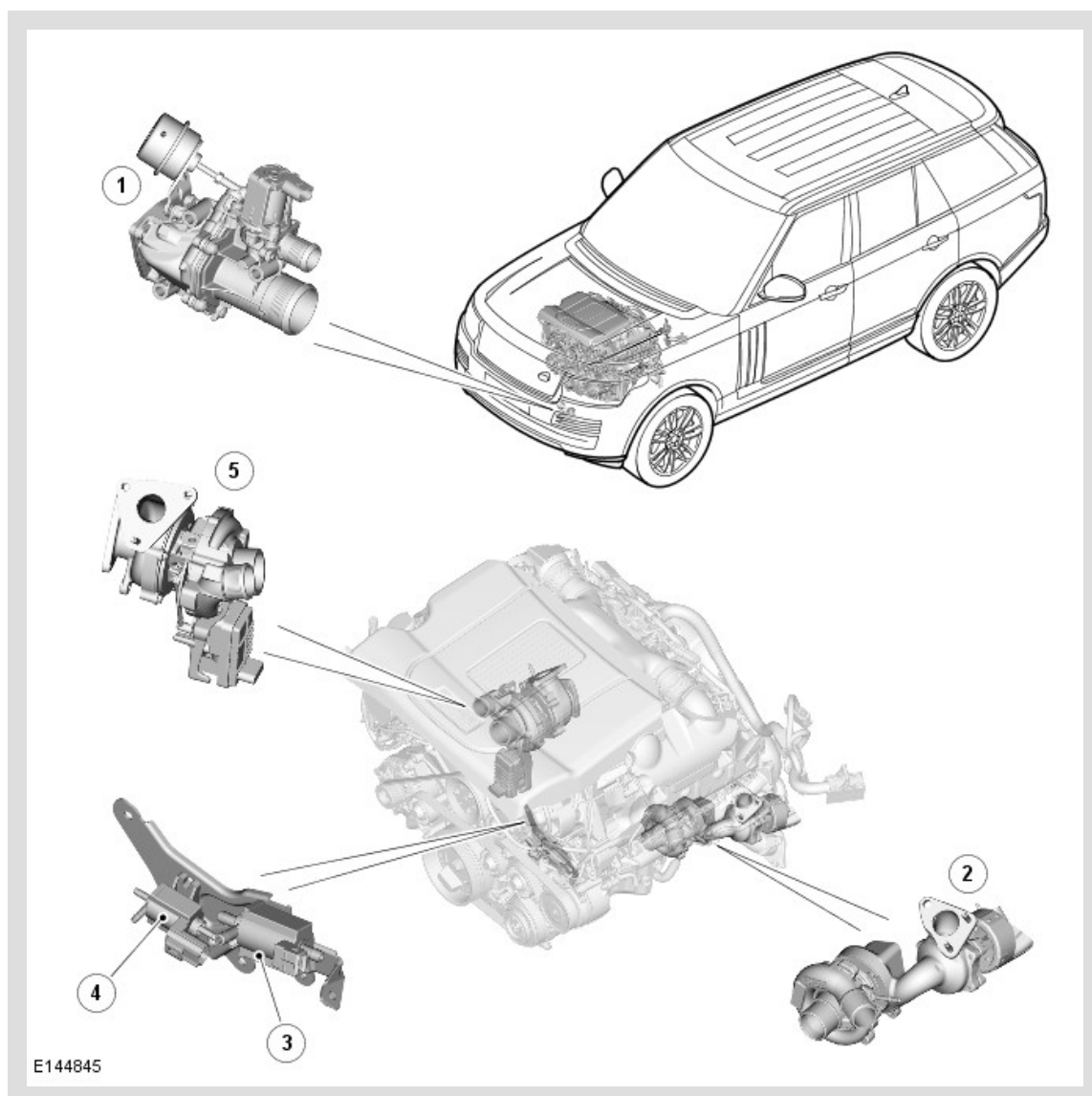


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2015.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - TURBOCHARGER - TDV8 4.4L DIESEL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Charge air intake valve
2	Secondary turbocharger
3	Turbine intake shut-off valve solenoid
4	Charge air solenoid
5	Primary turbocharger

OVERVIEW

The 4.4L V8 diesel engine uses two turbochargers, a VGT (variable geometry turbine) type (the primary turbocharger) and a fixed vane type (the secondary turbocharger).

Both turbochargers are used in a parallel sequential turbocharging system which enables the engine to achieve quick throttle response at low engine speeds and efficient use of exhaust gas energy at high engine speeds. The primary turbocharger operates through the entire engine speed range, but is at its most efficient at engine speeds of up to 2400 rev/min, running in a mono turbocharger mode. At engine speeds above 2400 rev/min under load, the secondary turbocharger comes into operation, with both of the turbochargers running in a parallel bi-turbocharger mode.

The parallel sequential turbocharging system comprises the two turbochargers, a charge air intake valve, a turbine intake shut-off valve solenoid and a charge air solenoid. Operation of the system is controlled by the ECM (engine control module) .

The turbochargers and the charge air intake valve are interconnected by the charge air ducts of the intake air distribution and filtering system.

For additional information, refer to: [Intake Air Distribution and Filtering](#) (303-12E Intake Air Distribution and Filtering - TDV8 4.4L Diesel, Description and Operation).

DESCRIPTION

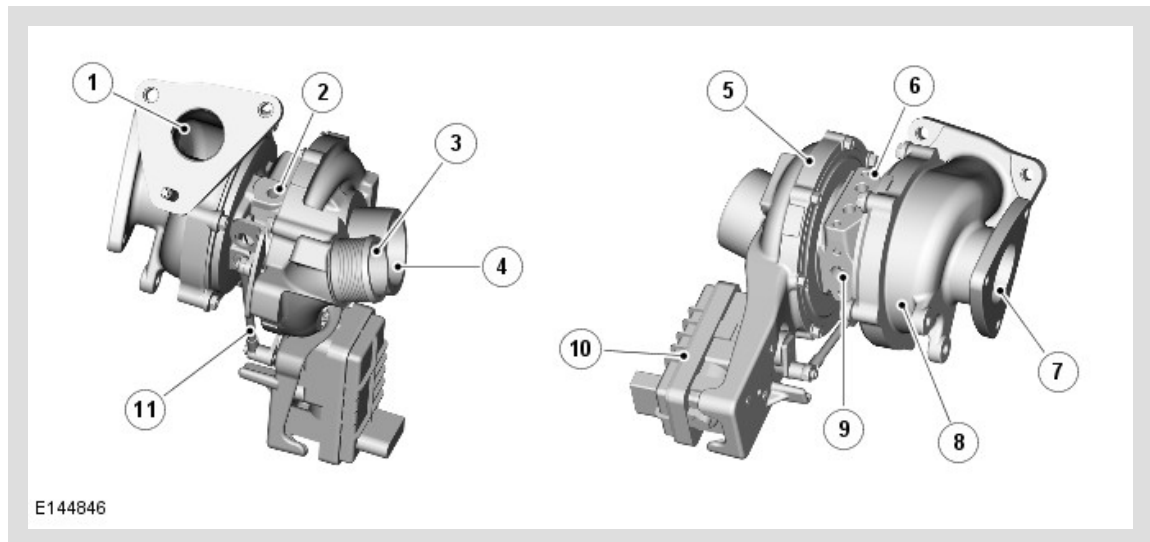
TURBOCHARGERS

Each turbocharger consists of two turbo elements, a turbine wheel and compressor wheel, enclosed separately in cast housings and mounted on a common shaft, which rotates in two semi-floating bearings.

The turbocharger bearings are lubricated and cooled by an oil supply from the cylinder block. The return oil drains into a dedicated area of the oil pan, assisted by an oil scavenge element in the vacuum pump.

For additional information, refer to: [Engine](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2 /TDV6 3.0L Diesel - Gen 1.5, Description and Operation).

PRIMARY TURBOCHARGER



ITEM	DESCRIPTION
1	Exhaust intake
2	Oil feed connection
3	Charge air outlet
4	Clean air intake
5	Compressor housing
6	Center housing
7	Exhaust outlet
8	Turbine housing
9	Oil return connection
10	VGT vane actuator
11	Actuator rod

The primary turbocharger is attached to the right exhaust manifold and is secured to a flange on the manifold with three studs and nuts. A second flange on the turbocharger provides for the attachment of the right down pipe of the exhaust system. Three screws secure the down pipe to the flange and a gasket seals the joint.

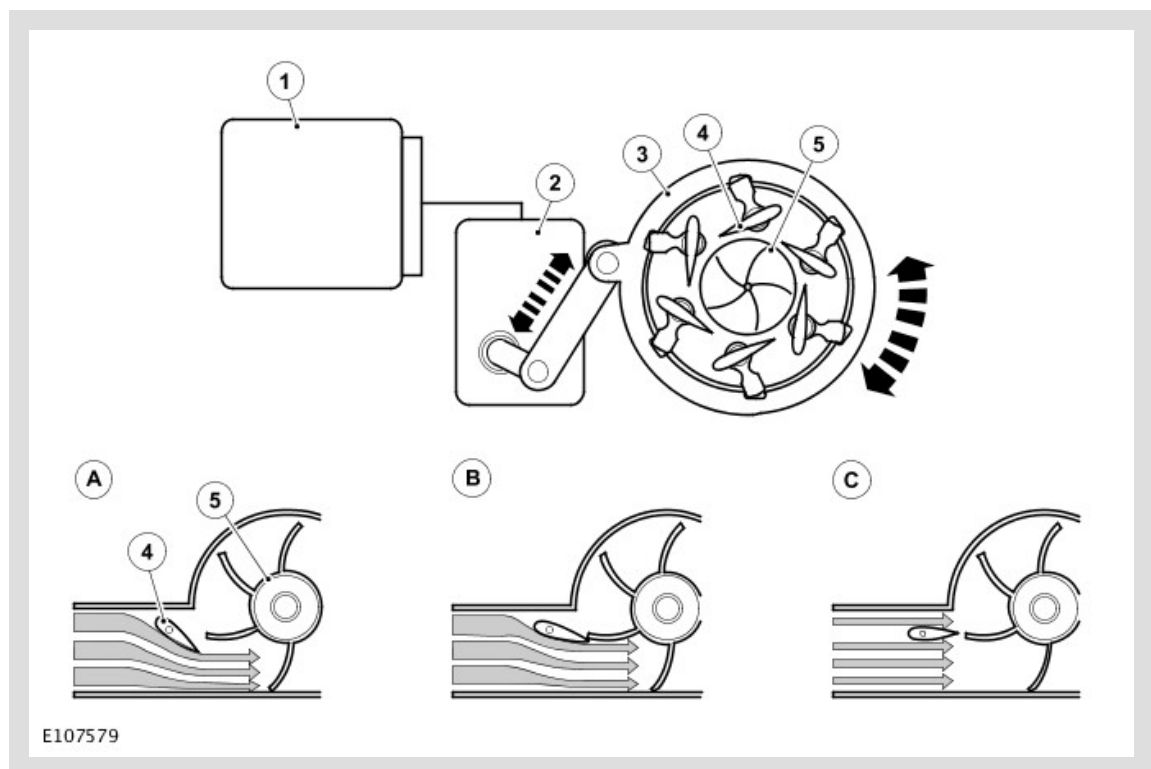
The compressor end of the turbocharger has two hose connections. The central hose connection provides a clean air supply from the air cleaner to the compressor. The second connection on the outside of the housing provides for the attachment of the charge air ducts.

An integral bracket houses the VGT vane actuator. The actuator is connected to an eccentric lever which moves the turbine housing to adjust the position of the vanes. When the actuator operates a boss is rotated, which in turn moves the lever and changes rotary motion into linear motion. The lever is connected to a rod attached to the outside of the turbine housing and the linear motion is converted back to rotary motion of the housing.

Operation of the VGT vane actuator is controlled by the ECM . The VGT vane actuator also provides the ECM with a feedback signal to determine the pitch angle of the vanes.

The variable vanes in the turbocharger improve the exhaust gas power transfer to the turbine wheel which in turn drives the compressor wheel. At low engine speeds this greatly assists the increase in charge air pressure.

Principles of Variable Vane Operation



A = LOW ENGINE SPEED; B = INTERMEDIATE ENGINE SPEED; C = MAXIMUM ENGINE SPEED.

ITEM	DESCRIPTION
1	Engine control module
2	VGT vane actuator

3	Turbine housing
4	Variable vanes
5	Compressor wheel

Low Engine Speed. At low engine speeds the volume of exhaust gas is low so the vanes are moved towards the closed position to reduce the turbine inlet area. This reduction causes an increase in the gas velocity into the turbine wheel thereby increasing wheel speed and charge air pressure.

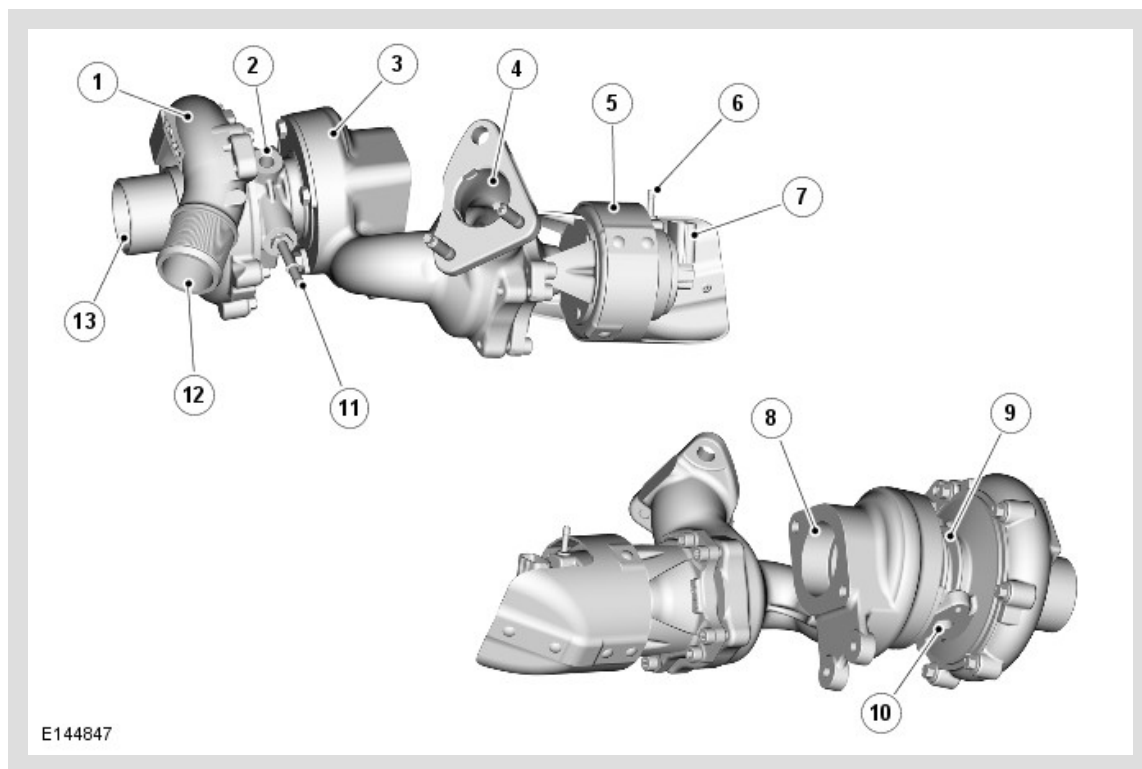
Intermediate Engine Speed. As the engine speed increases and the volume of exhaust gas increases the vanes are moved towards the open position to increase the turbine inlet area and maintain the gas velocity.

Maximum Engine Speed. At maximum engine speed the vanes are almost fully open maintaining the gas velocity into the turbine wheel.

When the vehicle is driven at high altitudes the ambient pressure reduces causing the compressor wheel to rotate faster to achieve the same charge air pressure. To prevent the turbine wheel from over-speeding under these conditions, the ECM protects the turbocharger by opening the vanes further to reduce the turbine wheel speed. This is known as the altitude margin of the turbocharger, which occurs at altitudes above 2000 meters. Below 2000 meters altitude compensation is not necessary. The ECM uses an internal barometric pressure sensor to monitor altitude.

The maximum position of the turbocharger vanes (fully open) is also an emergency default position in the event of an electrical fault. This lowers the chance of engine damage due to excessive charge air pressure.

SECONDARY TURBOCHARGER



ITEM	DESCRIPTION
1	Compressor housing
2	Oil feed connection
3	Turbine housing
4	Exhaust intake
5	Turbine shut-off valve
6	Vacuum connection
7	Position sensor electrical connector
8	Exhaust outlet
9	Center housing
10	Oil return connection
11	Seal pressurizing air pipe
12	Charge air outlet
13	Clean air intake

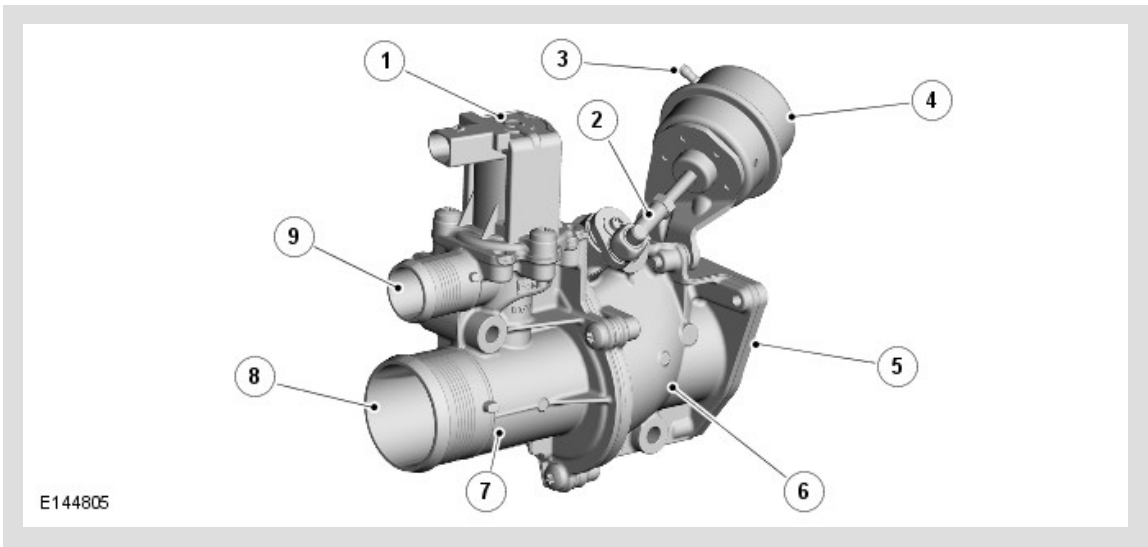
The secondary turbocharger is attached to the left exhaust manifold and is secured to a flange on the manifold with three studs and nuts. A second flange on the turbocharger provides for the attachment of the left down pipe of the exhaust system. Two studs and nuts secure the down pipe to the flange and a gasket seals the joint.

The compressor end of the turbocharger has two hose connections. The central connection provides a clean air supply from the air cleaner to the compressor. The second connection on the outside of the housing provides for the attachment of the charge air ducts.

A pipe on the center housing of the turbocharger provides for the connection of a bleed air pipe from the charge air duct connection with the electric throttle. The bleed air is used to pressurize the seals of the compressor stage to prevent oil leaking into the compressor housing when the turbocharger is not running.

Attached to the rear of the turbocharger is a turbine shut-off valve operated by a vacuum actuator. The ECM operates the turbine intake shut-off valve solenoid to apply and release the vacuum in the actuator, which opens the turbine shut-off valve when a vacuum is present. The turbine shut-off valve is closed when the system is operating in mono-turbocharger mode, diverting exhaust gasses from the left exhaust manifold, via the exhaust crossover duct to the right exhaust manifold. A position sensor in the turbine shut-off valve allows the ECM to monitor the shut-off valve for correct operation.

CHARGE AIR INTAKE VALVE



ITEM	DESCRIPTION
1	Recirculation solenoid

2	Shut-off valve connecting rod
3	Vacuum connection
4	Vacuum actuator
5	Shut-off valve charge air outlet (to charge air coolers)
6	Shut-off valve body
7	Recirculation solenoid valve body
8	Charge air intake (from secondary turbocharger)
9	Recirculation solenoid charge air outlet (to primary turbocharger clean air duct)

The charge air intake valve is attached to a bracket on the front subframe and the cooling pack protector. The ECM uses the valve to control the bi-turbocharger operation of the turbochargers.

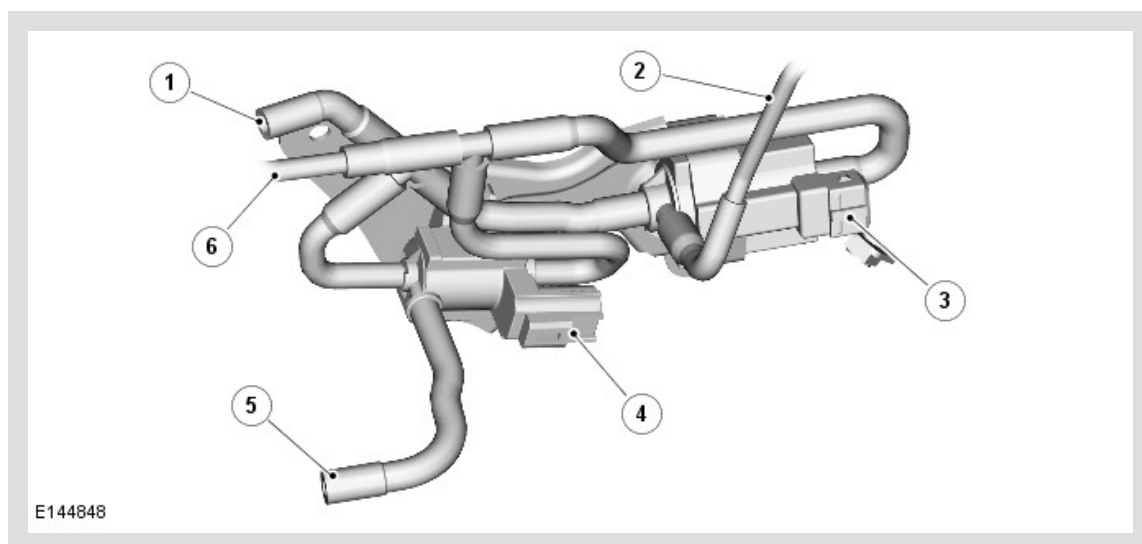
The charge air intake valve consists of a recirculation solenoid and a shut-off valve that are connected together by four screws and a gasket. The recirculation solenoid controls the flow of air from the compressor of the secondary turbocharger to the clean air duct of the primary turbocharger. The shut-off valve controls the flow of air from the compressor of the secondary turbocharger to the charge air coolers.

The recirculation solenoid is operated by a power feed from the EJB (engine junction box) and a ground connection with the ECM . When it is de-energized, the recirculation solenoid is closed.

If the recirculation solenoid develops a fault it will default to the closed position. In this position an overspeed of the primary turbocharger will occur. This overspeed is sensed by the MAP (manifold absolute pressure) sensor which detects the increase in pressure. The ECM registers the signals from the MAP sensor and operates the turbochargers in mono-turbo mode with restricted engine torque and DTC (diagnostic trouble code) 's recorded.

The shut-off valve is a flap valve operated by a vacuum actuator. The ECM operates the charge air solenoid to apply and release the vacuum in the actuator, which opens the valve when a vacuum is present. When the shut-off valve is open charge air from the secondary turbocharger flows through the valve to the charge air coolers.

VACUUM CONTROL SOLENOIDS



ITEM	DESCRIPTION
1	Pipe from vacuum pump
2	Pipe to turbine shut-off valve (secondary turbocharger)
3	Turbine intake shut-off valve solenoid
4	Charge air solenoid
5	Pipe to vacuum actuator of charge air shut-off valve
6	Pipe from left air cleaner

The turbine intake and charge air solenoids are installed on a bracket attached to the front of the left cylinder head. The ECM operates the solenoids to apply and release the vacuum in the vacuum actuators of the turbine shut-off valve (of the secondary turbocharger) and the charge air shut-off valve.

The vacuum for the actuators is produced by the vacuum pump on the rear of the right cylinder head.

For additional information, refer to: [Engine](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2 /TDV6 3.0L Diesel - Gen 1.5, Description and Operation).

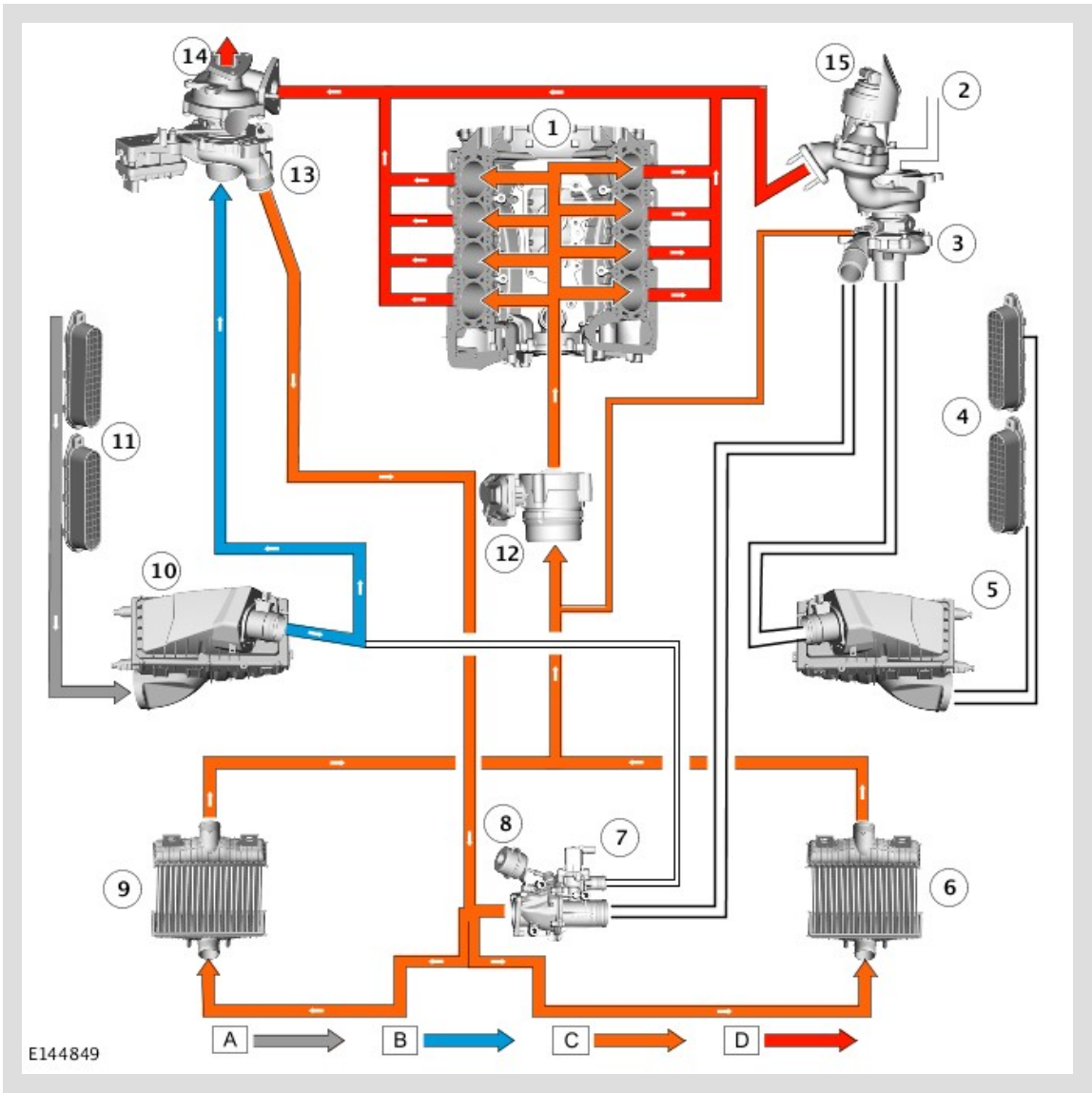
The solenoids are normally-closed valves installed in the vacuum line between the vacuum reservoir and their related vacuum actuator. Each solenoid is also connected to a line from the left air cleaner, which allows air into the vacuum lines to the actuators when the solenoids are de-energized.

OPERATION

The turbine wheel of each turbocharger uses the engine exhaust gasses to drive the compressor wheel. The compressor wheel draws in clean air, which is supplied to the engine cylinders in a compressed form.

The turbochargers can operate in two modes; mono turbocharger operation or bi-turbocharger operation.

MONO TURBOCHARGER OPERATION



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASSES.

ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Secondary turbocharger
4	Left intake grills
5	Left air cleaner

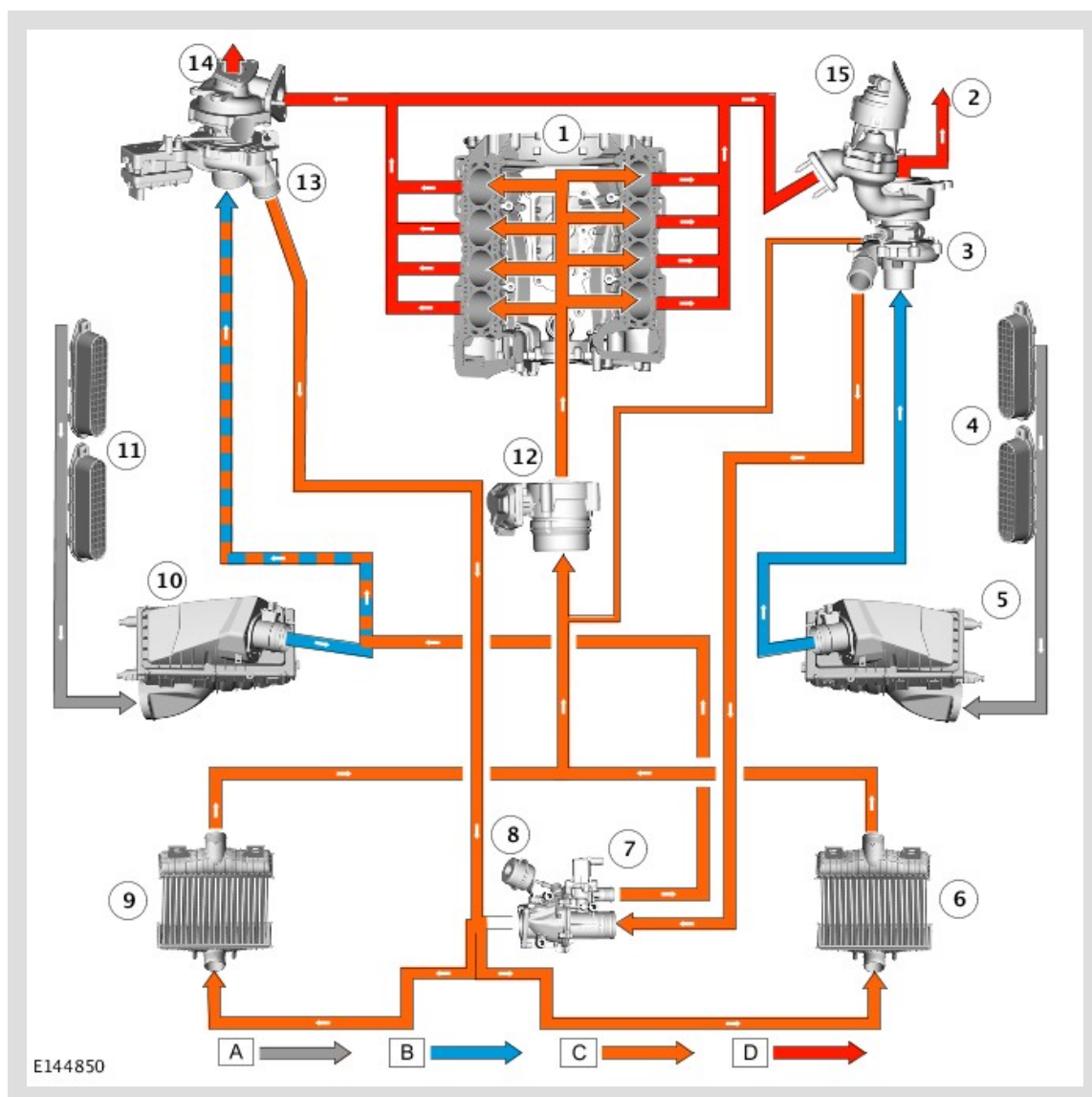
6	Left charge air cooler
7	Charge air recirculation solenoid
8	Charge air shut-off valve
9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Primary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

Fresh air is drawn through the right air cleaner and the MAF (mass air flow) sensor to the primary turbocharger compressor. The compressed air then passes through the charge air coolers and electric throttle to the intake manifolds.

The turbine shut-off valve on the secondary turbocharger is closed and therefore exhaust gasses are unable to operate the secondary turbocharger turbine. In this condition all charge air pressure is produced by the primary turbocharger using the exhaust gasses from all eight cylinders.

If the engine idles for more than 3 minutes, the secondary turbocharger is actuated to keep the turbocharger shaft rotating, to prevent oil from leaking into the turbine housing. This is achieved by periodically opening the turbine shut-off valve to operate the turbocharger.

BI-TURBOCHARGER SWITCHING



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASSES.

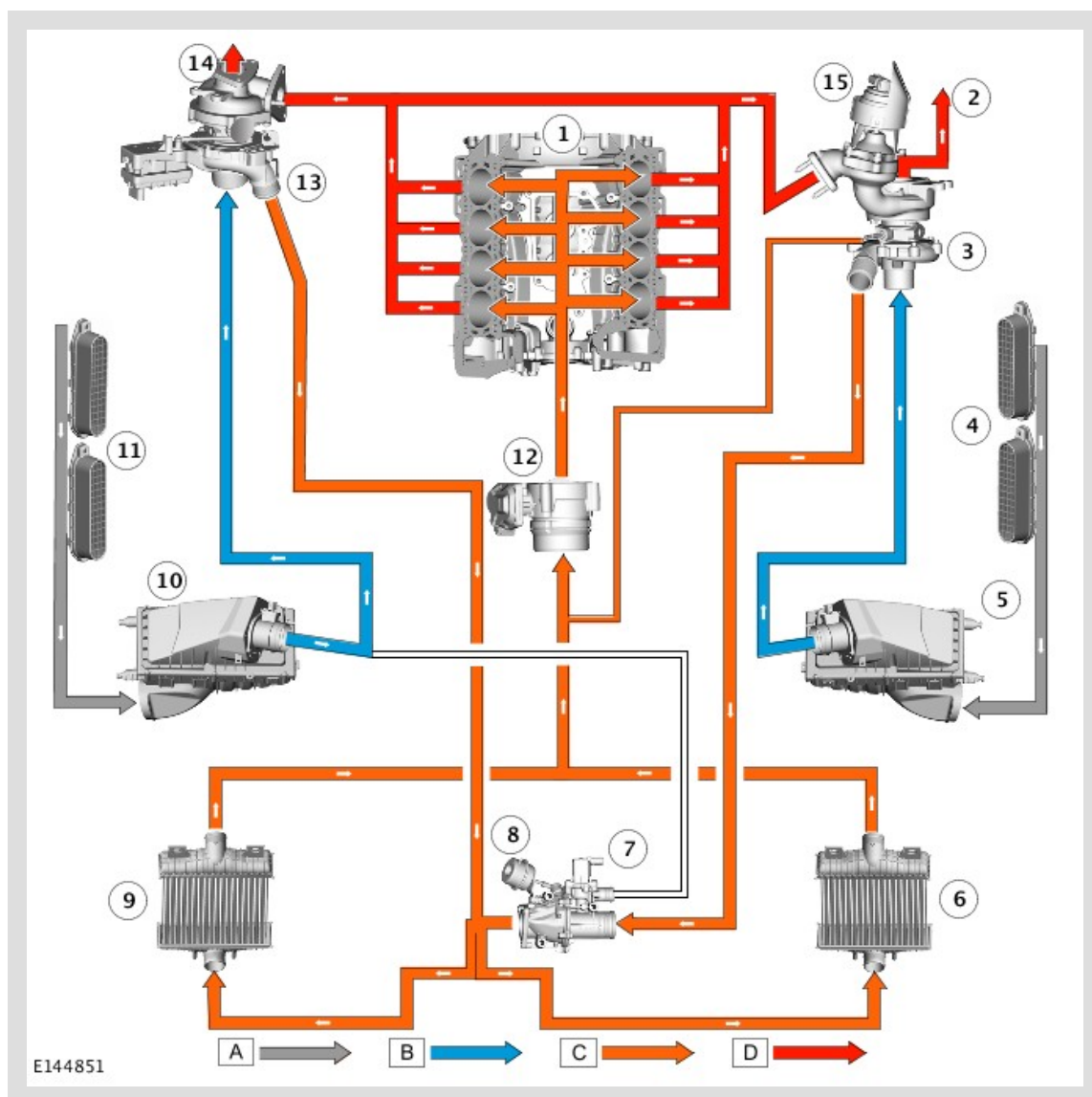
ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Primary turbocharger
4	Left intake grills

5	Left air cleaner
6	Left charge air cooler
7	Charge air shut-off valve
8	Charge air recirculation solenoid
9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Secondary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

When the engine operating parameters approach the limits (approximately 2800 rev /min under load) of the primary turbocharger, the ECM begins the switch to parallel bi-turbocharger operation. The secondary turbocharger is brought into operation by the opening of the turbine shut-off valve which allows exhaust gasses to flow through the turbine. Fresh air is then drawn through the left air cleaner and the MAFT (mass air flow and temperature) sensor to the secondary turbocharger compressor.

Initially, the secondary turbocharger does not produce a charge air pressure to equal that of the primary turbocharger. Therefore, the charge air from the secondary turbocharger is fed via the recirculation solenoid into the clean air duct of the primary turbocharger.

BI-TURBOCHARGER OPERATION



A = DIRTY AIR; B = CLEAN AIR; C = CHARGE AIR; D = EXHAUST GASSES.

ITEM	DESCRIPTION
1	Engine
2	Exhaust system left intake pipe
3	Primary turbocharger
4	Left intake grills

5	Left air cleaner
6	Left charge air cooler
7	Charge air shut-off valve
8	Charge air recirculation solenoid
9	Right charge air cooler
10	Right air cleaner
11	Right intake grills
12	Electric throttle
13	Secondary turbocharger
14	Exhaust system right intake pipe
15	Turbine shut-off valve

As the secondary turbocharger charge air pressure increases, the recirculation solenoid is closed and the charge air shut-off valve opened. This increases the charge air pressure from the secondary turbocharger and combines it with the charge air from the primary turbocharger.

The ECM maintains the turbochargers in bi-turbocharger mode until it determines that the engine operating parameters no longer require it. The ECM then switches the turbochargers back to mono turbocharger operation.

SHUT-OFF VALVE CONTROL

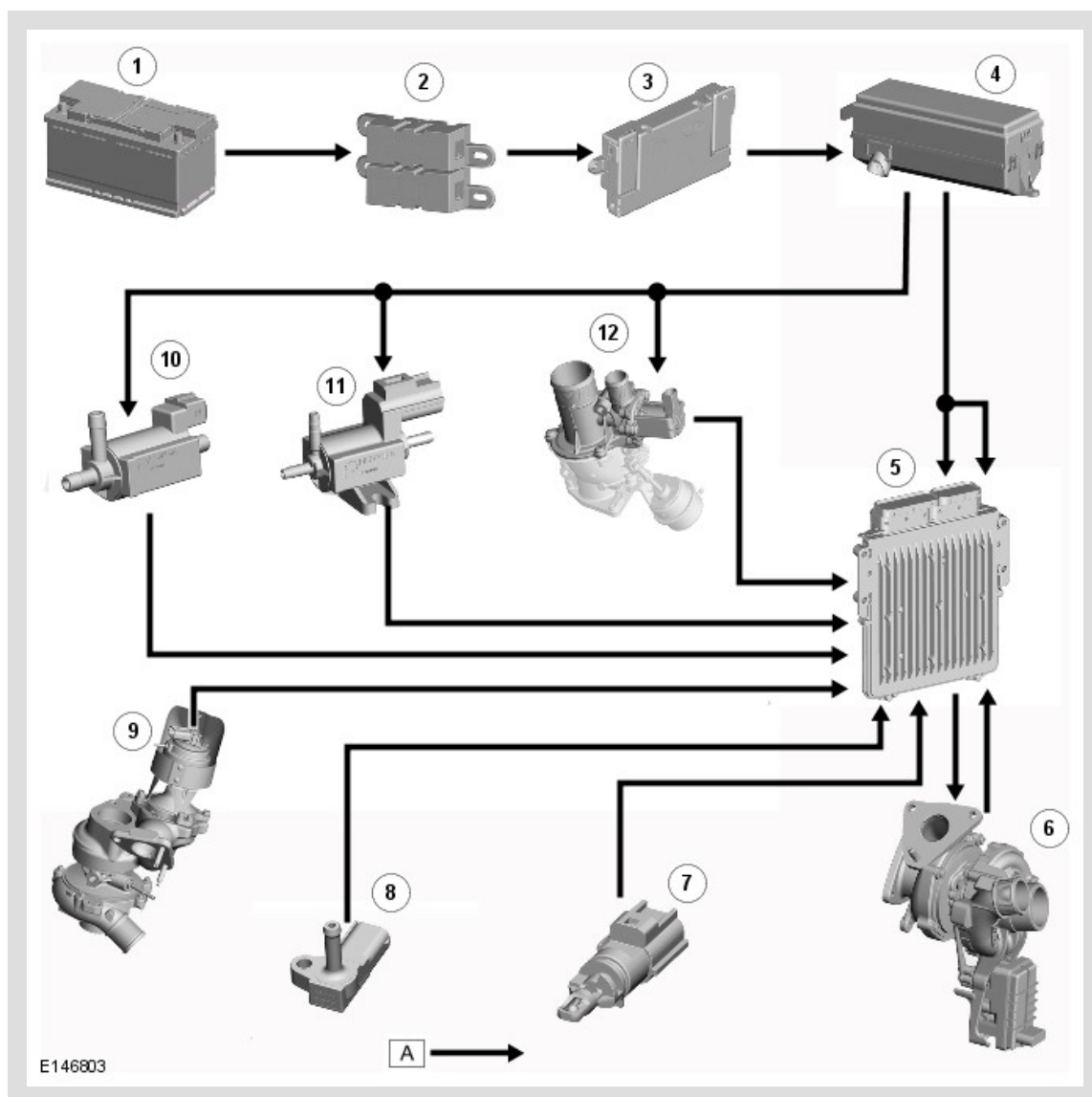
SECONDARY TURBOCHARGER TURBINE INTAKE SHUT-OFF VALVE

If vacuum is lost to the shut-off valve it will default to the closed position. The turbine intake valve position sensor will inform the ECM , which will initiate mono-turbo mode, restrict engine performance and record a DTC .

CHARGE AIR SHUT-OFF VALVE

If vacuum is lost to the shut-off valve it will default to the closed position. In the default position, the ECM initiates mono-turbo operation and restricts engine torque. A DTC will also be recorded in the ECM .

CONTROL DIAGRAM



A = HARDWIRED.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Engine junction box

5	Engine control module
6	VGT vane actuator
7	Secondary turbocharger exhaust temperature sensor
8	Charge air pressure sensor
9	Turbine shut-off valve position sensor
10	Turbine intake shut-off valve solenoid
11	Charge air solenoid
12	Charge air recirculation solenoid